D 30585 **(Pages : 2)** **Name.....**

Reg. No.....

FIFTH SEMESTER (CBCSS—UG) DEGREE EXAMINATION
NOVEMBER 2022

Physics/Applied Physics

PHY 5B 07/APH 5B 07—QUANTUM MECHANICS

(2019 Admission onwards)

Time : Two Hours Maximum : 60 Marks

The symbols used in question paper have their usual meanings.

Section A (Short Answer Type)

Answer **all** *questions in two* or *three sentences.*
Each correct answer carries a maximum of 2 *marks.*

1. Plot the radiancy versus wavelength spectrum of a blackbody for two different temperatures.
2. What is the origin of the unmodified peak in the Compton spectrum ?
3. List any four basic properties of photons.
4. Explain Ritz combination principle.
5. What was the purpose of Davisson-Germer experiment ?
6. Explain the principle of complementarity.
7. Explain how the amplitude of de Broglie wave related to the probability of finding the particle.
8. Discuss the probability interpretation of wavefunction.
9. Give an expression for the energy eigen values of a one-dimensional simple harmonic oscillator. How is it different from that of a particle in a box, in terms of the spacing ?
10. Discuss the ammonia inversion problem using quantum mechanical tunneling.
11. Explain the result of Stern-Gerlach experiment using a suitable example.
12. What is Bohr magneton ? What is its value ?

(Ceiling 20)

Turn over

Section B (Paragraph/Problem Type)

Answer **all** *questions in* **a paragraph of about half a page to one page**.
Each correct answer carries a maximum of 5 *marks.*

13. X-rays of wavelength 0.24 mm are Compton-scattered and the scattered beam is observed at an angle 60 degree relative to the incident beam. Determine :

    (a) The wavelength of the scattered X-rays ; and

    (b) Energy of the scattered X-rays.

14. Discuss the basic properties of atoms.

15. Calculate the de Broglie wavelength of an electron having a kinetic energy 1 eV.

16. An electron moves in the x direction with a speed of 3.6×10^6 m/s. We can measure its speed to a precision of 1 %. With what precision can we simultaneously measure its position ?

17. Determine the average value of the position of a particle trapped in a one dimensional box of length L.

18. What are the quantum numbers associated with an atomic electron and what are their possible values ?

19. Explain normal and anomalous Zeeman effects.

(Ceiling 30)

Section C (Essay Type)

Essays.
Answer in about **two pages,** *any* **one** *question, correct answer carries* 10 *marks.*

20. Discuss the Rutherford nuclear atom model.

21. Setup the time-independent Schrödinger equation in one dimension.

(1 × 10 = 10 marks)